# Supplement to: Cross-Lagged Panel Networks

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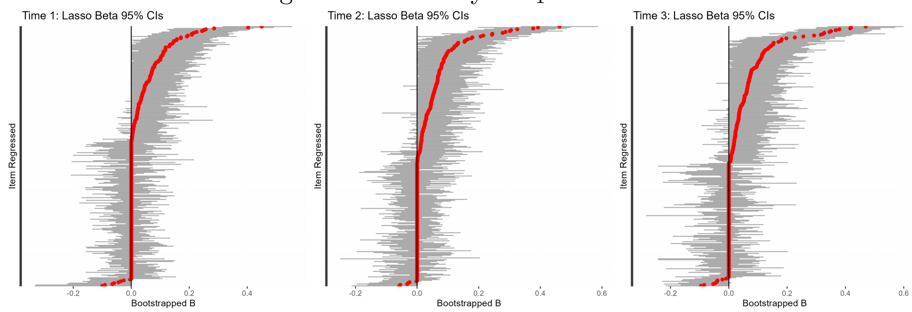
## 1 Bootstrap Results Demonstrating Stability of Network Structure Estimates

Due to the hybrid nature of the $\ell_1$-SEM approach, it is relevant to investigate the reproducibility of the $\ell_1$-derived model structure from the first step. Not only will this provide reproducibility, but it also gives more evidence towards the $\ell_1$-SEM and SEM approaches being interchangeable. To investigate this, we utilized a bootstrapping procedure with 1000 bootstraps, taking a full-sized random sample with replacement of the original data from the empirical example. Figure S1 shows the bootstrap results: The black vertical line indicates zero, red points indicate the regularized edge estimates from the initial estimation step, and the horizontal gray bars indicate the range within which 95% of bootstrapped edge estimates fall. Examining first the edges for which the sample estimate was zero (red dots along the black line) indicates that we do not have sufficient evidence to say that the sample zeros are nonzero across all time points. This is shown by the sample zeroes being well within their respective confidence intervals. Next, examining the confidence intervals of the nonzero estimates, these intervals largely fall to the same side of zero as the sample estimates, suggesting a greater assurance that these parameters are indeed nonzero. This pattern is especially apparent as time progresses. With these results, we can proceed with greater confidence that the first pruning step results in reproducible results.

## 2 Simulation Results: Convergence

The Panel VAR methodology encountered difficulties with convergence. Across both processes and all models, the median edge distances (MAE) ranged from 0.01-0.03. In the nonstationary process, many conditions resulted in much higher MAE values ranging as high as 595. Many of them came from the high density ($m = 0.8$) and high parameter ($p$= 16) conditions, but most conditions encountered issues, regardless of the population model. In the stationary pro-

Figure S1: Stability of $\ell_1$ Estimates




*Note.* Each graph represents the bootstrapped distribution of all parameter estimates across different time points. Each red dot represents the sample estimate. The gray bars represent a 95% Confidence Interval. Each line on the y-axis represents a unique edge.

cess, the troublesome MAE values ranged as high as 862. A higher proportion of these came from the high density ($m = 0.8$) and high parameter ($p = 16$) conditions as well. Anecdotally, we noted that when the whole simulation was repeated using new random number seeds, the degree and conditions at which these failures occurred varied greatly, indicating a strong effect of Monte Carlo sampling variability on convergence failures. Given the median ranges across all of the conditions, we excluded any Panel VAR results that had a mean edge distance higher than 1.0. The proportion of excluded samples are shown in Table 1.

Table 1: Panel VAR Convergence Failures

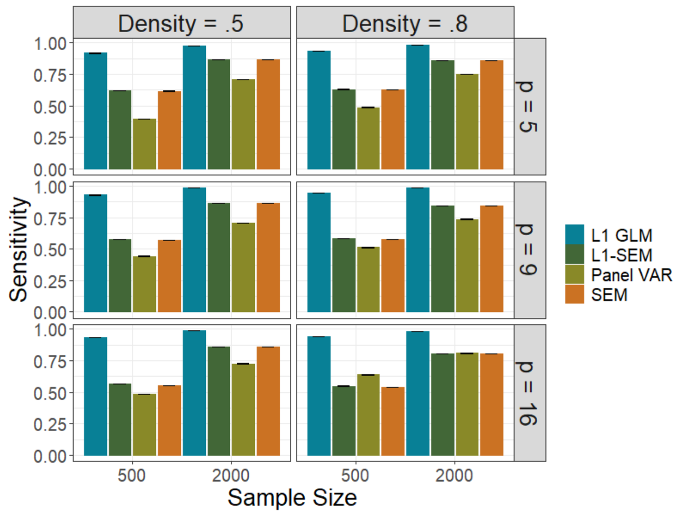
| | Nonstationary | | Stationary | |
|---|---|---|---|---|
| | $m = 0.5$ | $m = 0.8$ | $m = 0.5$ | $m = 0.8$ |
| $n = 500$ | | | | |
| $p = 5$ | 0.05% | 0.25% | 0.15% | 0.20% |
| $p = 9$ | 0.00% | 0.05% | 0.25% | 0.25% |
| $p = 16$ | 0.40% | 1.40% | 1.10% | 1.85% |
| $n = 2000$ | | | | |
| $p = 5$ | 0.05% | 0.00% | 0.00% | 0.05% |
| $p = 9$ | 0.10% | 0.45% | 0.30% | 0.25% |
| $p = 16$ | 0.25% | 1.30% | 0.70% | 2.95% |

*Note:* $n$ represents the sample size, $p$ represents the number of parameters, and $m$ represents the density

# 3 Simulation 2 (Stable Person-level Effects)

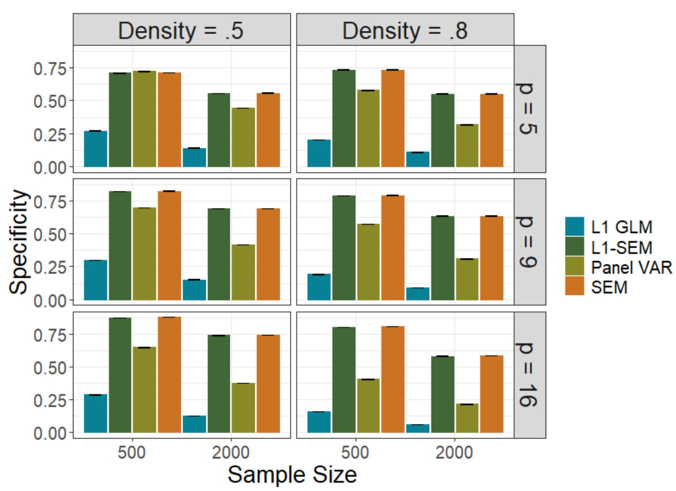
The second simulation is described in the paper. Figures depicting all results are shown here.

Figure S2: Sensitivity of Estimated Network Structure Under Nonstationarity Process with Stable Person Means



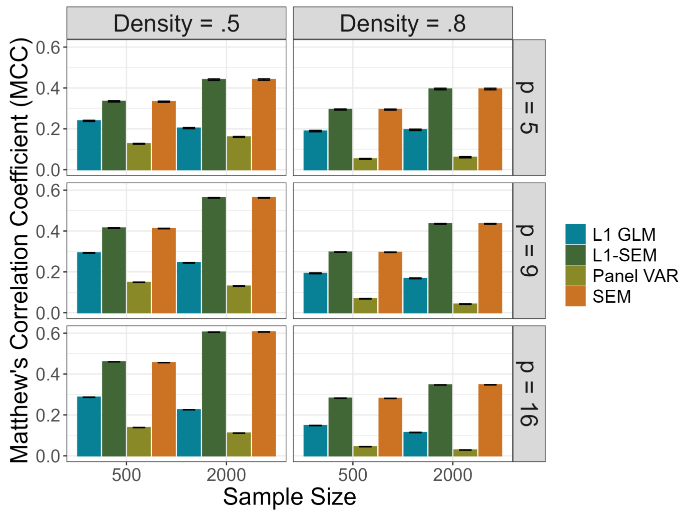

*Note:* Error bars represent ± one standard error from the mean

Figure S3: Specificity of Estimated Network Structure Under Nonstationarity Process with Stable Person Means



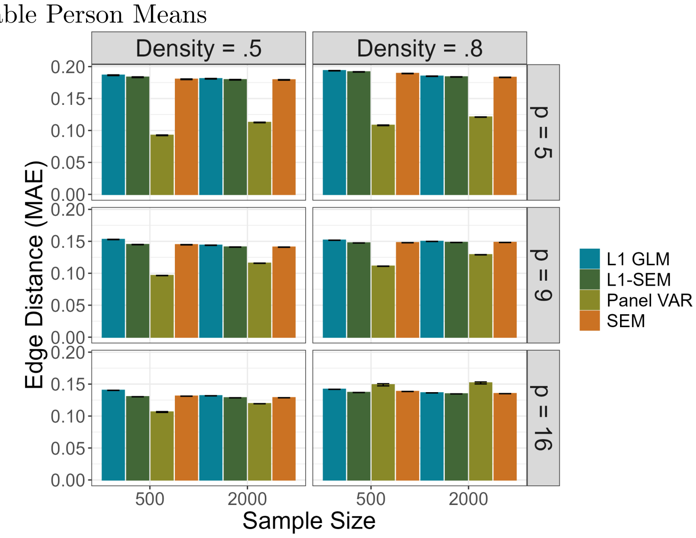

*Note.* Error bars represent ± one standard error from the mean.

Figure S4: Matthew's Correlation Coefficient Under Nonstationarity Process with Stable Person Means



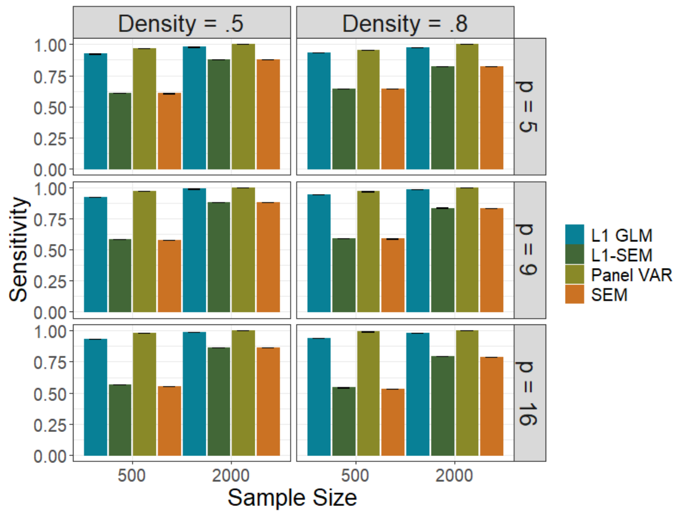

*Note.* Error bars represent ± one standard error from the mean.

Figure S5: Mean Absolute Error of Edge weights Under Nonstationarity Process with Stable Person Means



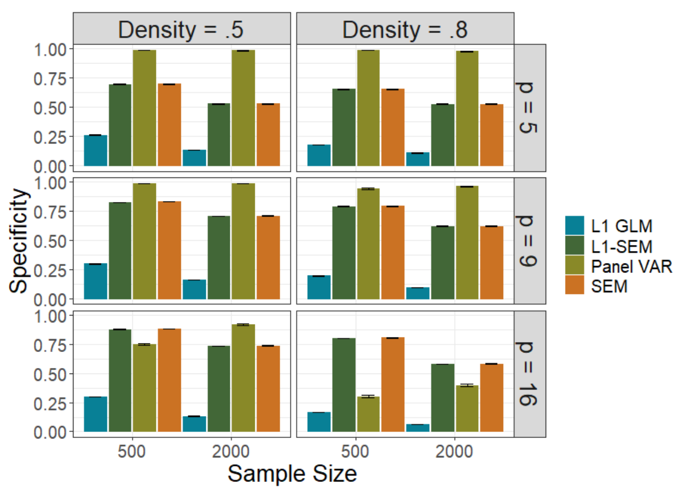

*Note.* Error bars represent ± one Monte Carlo standard error from the mean.

Figure S6: Sensitivity of Estimated Network Structure Under Stationarity Process with Stable Person Means



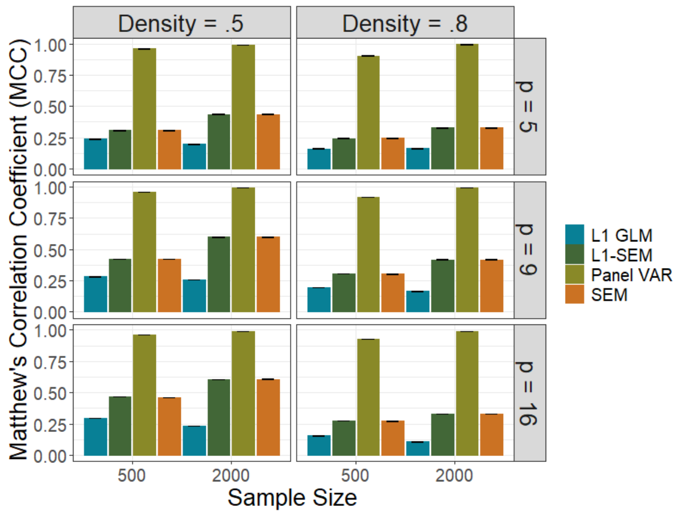

*Note.* Error bars represent ± one Monte Carlo standard error from the mean.

Figure S7: Specificity of Estimated Network Structure Under Stationarity Process with Stable Person Means



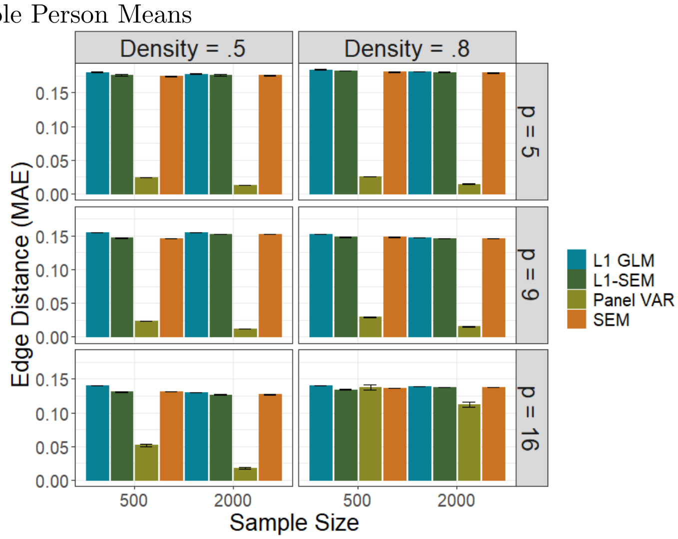

*Note.* Error bars represent ± one Monte Carlo standard error from the mean.

Figure S8: Matthew's Correlation Coefficient Under Stationarity Process with Stable Person Means




*Note.* Error bars represent ± one Monte Carlo standard error from the mean.

Figure S9: Mean Absolute Error of Edge weights Under Stationarity Process with Stable Person Means




*Note.* Error bars represent ± one Monte Carlo standard error from the mean.

Table 2: Panel VAR Convergence Failures: Stable Individual Differences

| | Nonstationary | | Stationary | |
|---|---|---|---|---|
| | $m = 0.5$ | $m = 0.8$ | $m = 0.5$ | $m = 0.8$ |
| $n = 500$ | | | | |
| $p = 5$ | 0% | 0% | 0% | 0% |
| $p = 9$ | 0% | 0% | 0% | 0.1% |
| $p = 16$ | 0.05% | 0.9% | 0.9% | 3.4% |
| $n = 2000$ | | | | |
| $p = 5$ | 0% | 0% | 0% | 0% |
| $p = 9$ | 0% | 0% | 0% | 0.25% |
| $p = 16$ | 0% | 0.6% | 0.05% | 2.85% |

*Note:* $n$ represents the sample size, $p$ represents the number of parameters, and $m$ represents the density

# 4 Item Text for Empirical Example

Table 3: Commitment to School and Self-Esteem Items Commitment to School

| Commitment to School | |
|---|---|
| SC1 | In general, I like school a lot. (R) |
| SC2 | School bores me. |
| SC3 | I don't do well at school. |
| SC4 | I don't feel like I really belong at school. |
| SC5 | Homework is a waste of time. |
| SC6 | I try hard at school. (R) |
| SC7 | I usually finish my homework. (R) |
| Self-Esteem | |
| SE1 | I feel that I'm a person of worth, at least on an equal level with others. (R) |
| SE2 | I feel that I have a number of good qualities. (R) |
| SE3 | All in all, I am inclined to feel that I'm a failure. |
| SE4 | I am able to do things as well as most other people. (R) |
| SE5 | I feel I do not have much to be proud of. |
| SE6 | I take a positive attitude toward myself. (R) |
| SE7 | On the whole, I am satisfied with myself. (R) |
| SE8 | I certainly feel useless at times. |
| SE9 | I wish I could have more respect for myself. |
| SE10 | At times I think I am no good at all. |

*Note.* Items marked with (R) were reverse-coded.